# Differential Geometry

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Lecture 1

## 1 Manifolds

The concept of a manifold is intended to generalise curves and surfaces in $\mathbb{R}^3$.

**Definition 1.1** (smooth structure)
A *smooth structure* on a topological space $M$ is a collection of *charts* (an *atlas*)

$$\varphi_\alpha : U_\alpha \to V_\alpha$$

where $U_\alpha \subseteq M$, $V_\alpha \subseteq \mathbb{R}^d$ are open, $\varphi_\alpha$ is a homeomorphism, such that

1. $M = \bigcup_\alpha U_\alpha$. That is, the charts cover $M$.
2. Given any charts $\varphi_\alpha, \varphi_\beta$, the map

$$\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \to \varphi_\beta(U_\alpha \cap U_\beta)$$

   is a smooth map between open subsets of $\mathbb{R}^d$.
3. If $\varphi$ is *compatible* (as in the previous point) with every chart in the atlas, then $\varphi$ is a chart in the atlas.

In practice (i.e. using Zorn's lemma), given any collection of charts satisfying the first two conditions, we can find a maximal atlas containing it.

**Remark 1.2.**
1. The second condition implies that $\varphi_\beta \circ \varphi_\alpha^{-1}$ is a diffeomorphism.
2. $d$ is fixed, assuming that $M$ is connected.

**Definition 1.3** (manifold)
A *(smooth) manifold* $M$ is a second countable Huasdorff space topological space, equipped with a smooth structure.
$d = \dim(M)$ is the *dimension* of $M$.

In practice, we may induce a topology on $M$ from a smooth structure (without the mention of open or homeomorphism), then $D \subseteq M$ is open if for any chart $\varphi_\alpha$, $\varphi_\alpha(D \cap U_\alpha)$ is an open subseteq of $\mathbb{R}^d$.

**Remark 1.4.**
1. We can also define a $C^k$ manifold by replacing smooth with $C^k$ as above. If $k = 0$, then we have a *topological manifold*.
2. On the other hand, we can replace $\mathbb{R}^d$ with $\mathbb{C}^n$, and obtain a *complex manifold*.

$\mathbb{R}^n$ is (trivially) a manifold.

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**Example 1.5** (unit sphere)

Define

$$S^n = \{x = (x_0, \ldots, x_n) \in \mathbb{R}^n \mid ||x|| = 1\} \subseteq \mathbb{R}^{n+1}$$

For the charts, we can consider the stereographic projections

$$\varphi(x) = \frac{1}{1 - x_0}(x_1, \ldots, x_n) \qquad \text{for } x_0 \neq 1$$

and

$$\psi(x) = \frac{1}{1 + x_0}(x_1, \ldots, x_n) \qquad \text{for } x_0 \neq -1$$

In particular,

$$\psi \circ \varphi^{-1}(u) = \frac{u}{||u||^2}$$

**Example 1.6**

If $M, N$ are manifolds of dimensions $d, e$ respectively, then $M \times N$ is a manifold of dimension $d + e$.

As a corollary, $T^n = (S^1)^n$ is a manifold.

**Example 1.7**

If $M$ is a manifold, $X \subseteq M$ is open, then $X$ is a manifold.